United International University

Department of Computer Science & Engineering

CSE 4326 — Microprocessors and Microcontrollers Laboratory

Real-Time Computer Vision & Weight-Based Produce Sorting System

Automated defect detection and gravimetric size grading
for agricultural supply chains — Bangladesh context

Group No.

6

Course Code

CSE 4326

Session

2024–2025

Platform

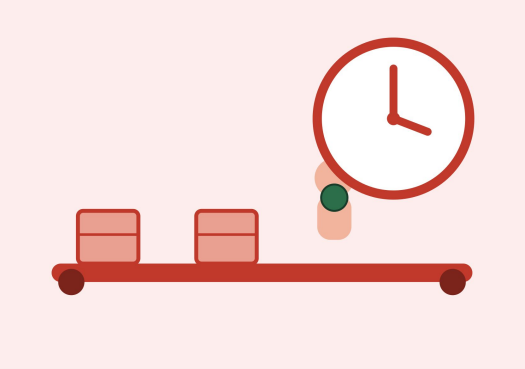
ESP32 + Raspberry Pi Zero 2W

Motivation & Problem Statement

Why does this project matter?

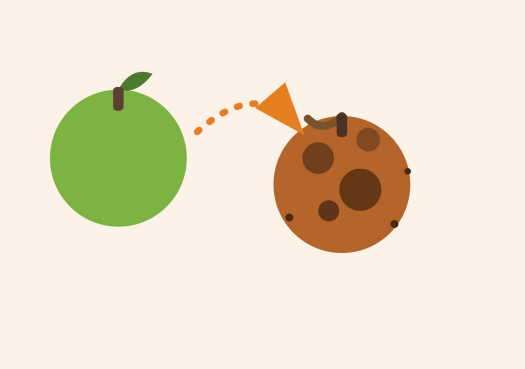
⚠️ Manual Sorting Bottlenecks

Traditional manual sorting processes only 40–60 kg/hr per worker. Human visual fatigue causes high error rates and missed defects — especially during peak harvest periods.



🦠 Post-Harvest Spoilage

Undetected surface rot and fungal lesions slip past manual inspection, contaminating healthy produce in bulk transport crates — leading to cascading losses downstream.



👷 Labor Scarcity in Rural Bangladesh

Rural agricultural regions face severe seasonal labor shortages during peak harvest periods. Affordable automation is essential for smallholder farmers and supply chain operators.



Project Introduction

What we are building and why it matters

This project designs and implements a tabletop automated produce sorting system that combines computer vision and gravimetric sensing on affordable embedded hardware. A Raspberry Pi5 performs real-time HSV color-based defect classification of produce items (Healthy / Rot / Green) using OpenCV, while an HX711 load cell assigns size grades (Small / Medium / Large) based on measured weight. Classification results are transmitted over UART to an ESP32 microcontroller, which computes timing delays and actuates two servo-driven diverter gates, physically directing items into one of four sorting bins. A local web dashboard streams live batch telemetry for operators, and an LED tower provides instant pass/fail feedback for non-technical users. The entire system targets rural smallholder farmers, cold-storage operators, and wholesale market traders in Bangladesh, replacing expensive industrial sorters with a sub-15,000 BDT open-hardware alternative.

2 **Processors**

Pi Zero 2W + ESP32

4 **Sorting Bins**

Reject / S / M / L

3 **Defect Classes**

Healthy / Rot / Green

6 **Week Build**

Full working prototype

Core System Features

Four major project capabilities

01 Computer Vision Inspection

- Pi Camera (CSI interface) captures item frames in real-time
- OpenCV HSV color thresholding classifies: Healthy / Rot-Damage / Green
- Runs entirely on Pi Zero 2W — no cloud, no GPU needed
- Classical CV approach: millisecond-scale per frame

02 Precision Weight-Based Grading

- HX711 24-bit ADC + load cell measures individual item mass
- Weight thresholds assign: Small / Medium / Large size class
- Integrated with vision result for combined sorting decision
- Compliant with commercial agricultural size standards

03 Environmental Telemetry

- DHT22 sensor monitors warehouse temperature & humidity
- Uses single-wire proprietary protocol (not I2C)
- Prevents bacterial decay via microclimate alerts
- Historical logs available via web dashboard

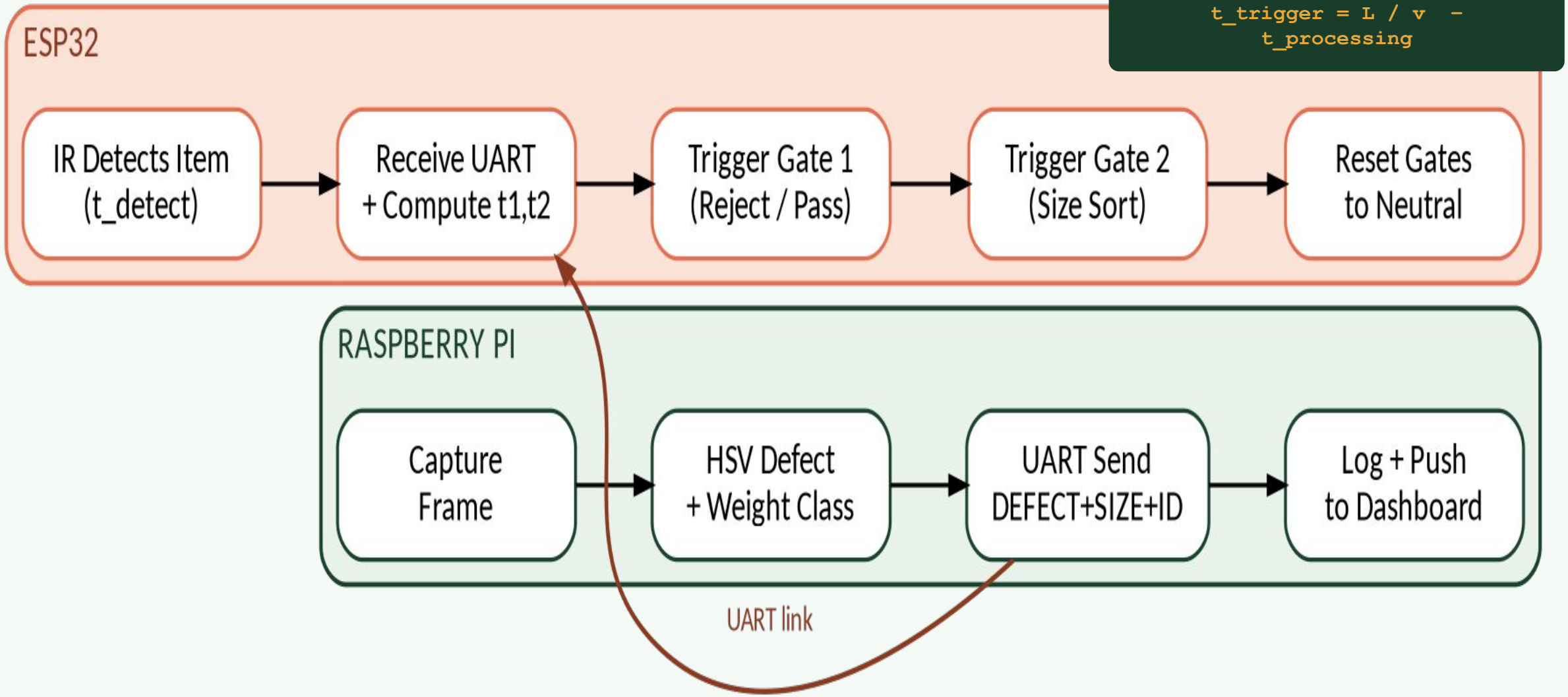
04 Wireless Web Dashboard & Alerts

- Flask/asyncio web server hosted on Raspberry Pi
- Streams live batch metrics, defect counts, telemetry
- LED tower provides instant visual pass/fail feedback
- Audio buzzer for non-technical farmer users

Firmware Flow — ESP32 & Raspberry Pi

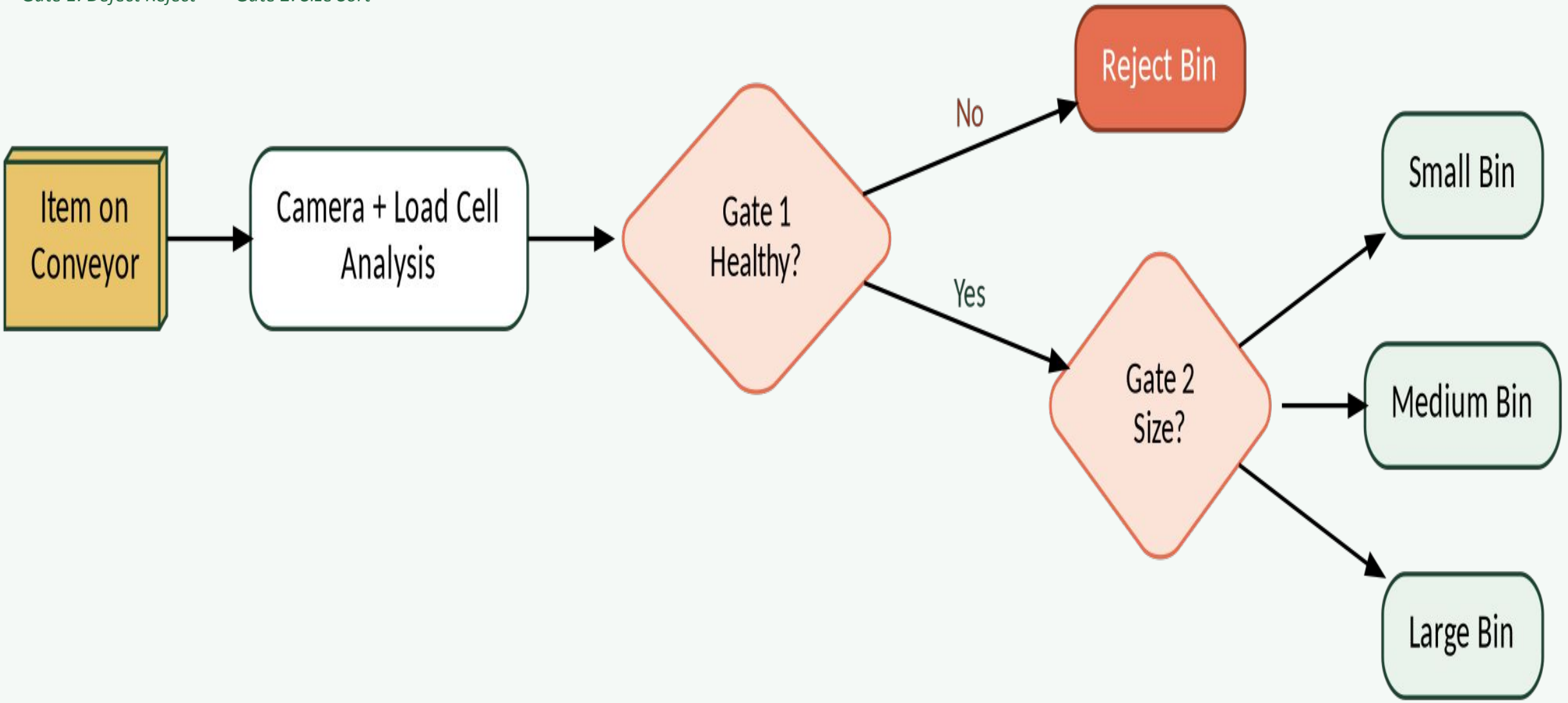
Synchronized dual-processor execution pipeline

```
t_trigger = L / v  -
t_processing
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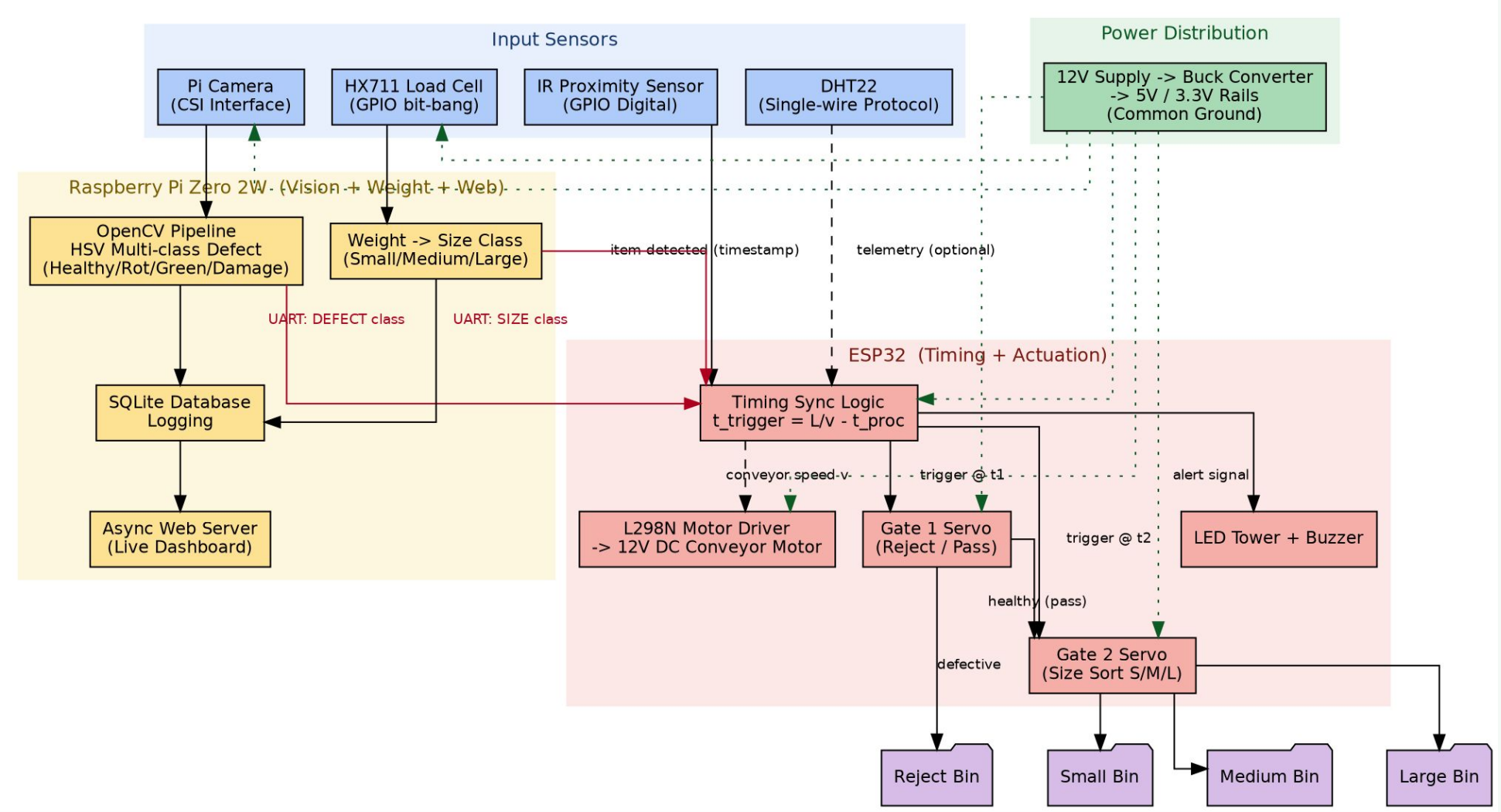
Two-Stage Sorting Logic

Gate 1: Defect Reject → Gate 2: Size Sort



Complete System Architecture — Reference Diagram

Full hardware, firmware, and data-flow map — sensors, power distribution, dual-processor pipeline, and actuation in one view



Hardware Components & Sensors

Apparatus required for full system implementation

PROCESSORS

- Raspberry Pi Zero 2W — Vision, DB, Web Server
- ESP32 DevKit V1 — Timing, Actuation, PWM

VISION & SENSING

- Pi Camera Module 5MP (CSI Interface)
- HX711 + Load Cell (1kg/5kg) — Weight Grading
- IR Proximity Sensor — Item Trigger
- DHT22 — Temperature & Humidity (1-wire)

POWER DISTRIBUTION

- LM2596 Buck Converter — 12V ⟶ 5V/3.3V
- 12V 3A DC Power Adapter — Main Supply

ACTUATION & MOTION

- 12V DC Gear Motor + L298N Driver — Conveyor Belt
- MG996R Servo Motor × 2 — Gate 1 (Reject) + Gate 2 (Size)

OUTPUT & FEEDBACK

- LED Tower / Multi-color LEDs — Status Indicator
- Active Buzzer — Audio Alert
- Flask Web Dashboard — Real-time Telemetry

STRUCTURAL

- Acrylic/PVC Frame + Mini Conveyor Belt
- 4× Physical Sorting Bins (Reject / S / M / L)
- Micro SD Card 32GB — Pi OS + SQLite DB
- Jumper Wires, Breadboard, Connectors

Three-Tier User Modularity

Designed for diverse real-world user profiles in Bangladesh

Profile 1

🌾 Illiterate Farmers

- Tactile push-button operation — no screen needed
- Vertical LED tower color signals (Green = Pass, Red = Reject)
- Audio buzzer cues for immediate sorting feedback
- Zero digital literacy required

Profile 2

🏭 Cold-Storage Operators

- Advanced local web application interface
- Continuous environmental logs (temp & humidity)
- Historical trend charts for storage monitoring
- Batch defect rate and size distribution reports

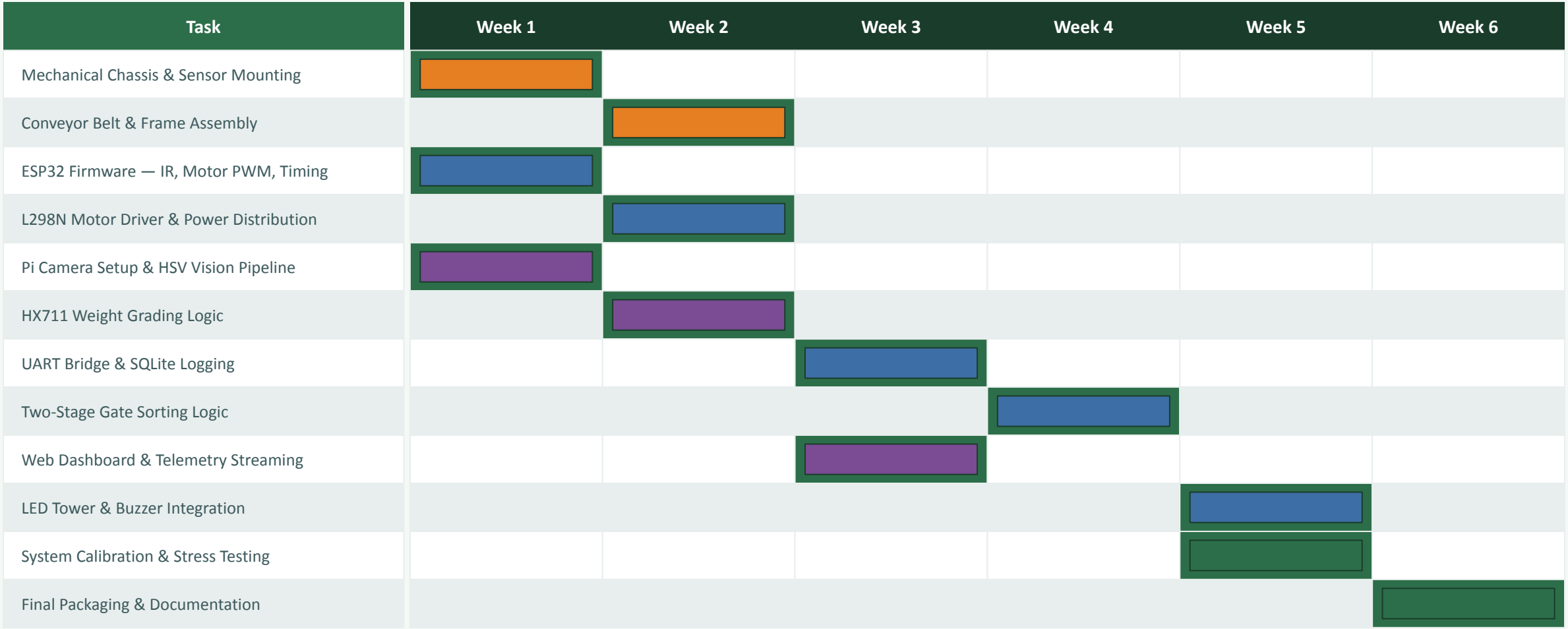
Profile 3

📊 Wholesale Traders (Aratdars)

- Standardized objective batch quality metrics
- Defect grading summaries per batch
- Transparent size-based valuation reports
- Eliminates buyer-seller disputes at market

Project Timeline — Gantt Chart

6-week implementation schedule — color-coded by team; tasks sharing a week run in parallel



Estimated Project Budget

Component-wise cost breakdown — all sourced locally in Bangladesh

#	Component	Qty	Unit (৳)	Total (৳)	Note
1	Raspberry Pi Zero 5	1	32000		Core Vision & Web Hosting
2	ESP32 Development Board (NodeMCU DevKit V1)	1	600	**600**	Real-time Timing & Actuation
3	Raspberry Pi Camera Module 5MP (CSI Cable)	1	1,400	**1,400**	Vision Inspection (CSI Port)
4	HX711 Module + Load Cell (1kg/5kg)	1	400	**400**	Weight-based Size Grading
5	IR Proximity Sensor Module	1	70	**70**	Item Trigger / Detection
6	DHT22 Temperature & Humidity Sensor	1	400	**400**	Telemetry (Single-wire)
7	L298N Motor Driver Module	1	250	**250**	12V Conveyor Motor Control
8	12V DC Gear Motor (Conveyor)	1	650	**650**	Conveyor Belt Motion
9	MG996R Servo Motor (Metal Gear)	2	500	**1,000**	Gate 1 (Reject) + Gate 2 (Size)
10	LM2596 DC-DC Buck Converter	1	150	**150**	12V ⟶ 5V Rail Regulation
11	12V 3A DC Power Adapter	1	500	**500**	Main Power Input
12	LED Tower + Active Buzzer	1 set	130	**130**	System Status Alerts
13	Micro SD Card 32GB Class 10	1	500	**500**	Pi OS & Local Database
14	Physical Structure (Acrylic, Belt, Bins)	1 set	2,200	**2,200**	Conveyor Frame & Sorting Bins
—	Misc (Jumper Wires, Breadboard, Connectors)	1 set	350	**350**	Wiring & Assembly
TOTAL ESTIMATED BUDGET				**৳ 10,400**	*Approximately BDT 13,400 — within university budget*

Sustainable Development Goal Alignment

Bangladesh SDG context — UN 2030 Agenda

2 SDG 2

Zero Hunger

Mitigates post-harvest food losses by automatically identifying and isolating contaminated produce before storage distribution. Reduces food waste at source — critical for Bangladesh where post-harvest losses reach 20–40% in perishables.

9 SDG 9

Industry, Innovation & Infrastructure

Integrates accessible edge computing (Raspberry Pi + ESP32) and sensor telemetry into agricultural supply chains. Provides a sub-15,000 BDT alternative to industrial sorters, making automation accessible to smallholder farmers.

12 SDG 12

Responsible Consumption & Production

Minimizes supply chain waste and establishes objective, standardized quality criteria for commercial agricultural trade. Eliminates subjective buyer-seller disputes and enables fair, transparent valuation of produce at wholesale markets.

Thank You

- ✔ Dual-processor pipeline — Pi Zero 2W + ESP32
- ✔ Classical HSV vision (Healthy / Rot / Green) — no GPU required
- ✔ Two-stage sorting: Gate 1 (Defect) → Gate 2 (Size)
- ✔ 4 physical bins: Reject / Small / Medium / Large
- ✔ t_trigger = L/v − t_processing timing sync
- ✔ Local web dashboard + LED/buzzer feedback
- ✔ Budget: ≈ ৳ 13,400 — affordable & locally sourced

Real-Time Computer Vision & Weight-Based Produce Sorting System

1. Introduction

Post harvest food loss in regional agricultural supply chains is caused by manual sorting problems, human fatigue and late detection of rot. Manual inspection techniques process few items per batch and can lead to grading errors. This project aims to develop an automated, cost-effective conveyor sorting system using optical computer vision, gravimetric mass scaling and environmental telemetry. The architecture is based on mathematical parameters to find defective products and optimize the management of the storage.

2. Problem Statement and Objectives

Manual sorting can only be done at 40-60 kg per hour per person, with great risk of error and the spread of microorganisms during storage, which cannot be monitored. The main goal is to develop a strong sorting machine, which replaces manual operation to inspect samples. The main design objectives include real-time defect detection, precise measurement of mass distribution, recording of microclimate data and structured access to the data for several tiers of users.

3. System Architecture and Hardware Integration

Physical assembly makes use of a strong and sleek frame for clean and expert tabletop assembly. It is fitted with a motorised rubber conveyor belt and an enclosed optical inspection tunnel. The electronic configuration features a dual-processor architecture that distributes real-time peripheral control from high-level computing.

Light sensors: Collects ambient light data.Motor: Controls motor action. It digitizes the pulses of an infrared proximity sensor at the belt inlet, records the weight signal data from a load cell platform, and gets the data of the environmental parameters measured by another sensor. It operates a DC gear motor and servo diverter gates.

Raspberry Pi: Performs higher level calculations. It captures images using a Pi Camera Module V2 inside the lit tunnel, executes OpenCV vision pipelines, maintains a local SQLite database and provides a local web dashboard.

The status feedback is provided by a vertical LED-based indicator tower, an integrated audio buzzer module and the wireless web interface.

4. Communication Protocols and Data Flow
The data communication between the elements within the system is provided by a number of specialized data communication buses, which ensure stability and speed.

I2C Bus: Connects the environmental sensors such as DHT22 with the processing unit.

SPI Bus: High speed camera module to the Raspberry Pi processor.

UART Serial Communication: Creates a hardware connection between the ESP32 microcontroller and Raspberry Pi, allowing the boards to execute commands simultaneously and trigger events.

5. Software and Computer Vision Implementation
The software framework is based on 4 functional features.

Computer Vision Inspection: The Raspberry Pi Camera takes repeated images inside the lit up tunnel. The matrix of thresholding is used to highlight surface lesions, fungal rot and dimensional anomalies in frames in the HSV space.

Precision Mass Grading: The load cell platform HX711 is used to measure continuous voltage changes in response to the weight of the item, allowing for categorization of commercial size.

Environmental Telemetry: DHT22 sensor to keep the parameters of ambient temperature and humidity under check to avoid bacterial and fungal growth in storage zones.

Wireless Web Dashboard: Is an asynchronous local web server that pushes real-time batch metrics, historical logs and system status alerts to any connected local network device.

6. Three-Tier User Modularity
The system has three user interfaces that can be adapted to a variety of operating situations.

Illiterate Farmers: Physical interaction, uses a tactile push button, a vertical tower with LED colour indicators and an audio buzzer for immediate verification of pass/fail sorting.

Cold-Storage Operators: Advanced web application interface that keeps logs of environment continuously, historical temperature, tracking inventory.

Wholesale Market Traders: Standardized metrics for batch quality, objective defect grading summaries and transparent valuation reports eliminate pricing disputes.

7. Project Time Line (Six Weeks)
The project timeline breaks the project into five engineering team members.

Week 1: Mechanical chassis building, conveyor belt assembling, and sensor mounting physically.

Week 2: Firmware development for ESP32 sensors, implementation of signal reading and setting up PWM control of motor.

Week 3: Raspberry Pi Camera setup, Implementation of OpenCV vision pipeline and calibration of color thresholding.

Week 4: Setting up UART communication bridge and configuring SQLite local database logging.

Week 5: Deploying a web dashboard, streaming of real-time telemetry and integration of actuators.

Week 6: System calibration, stress testing and final packaging.

8. Budget and Cost-Effectiveness
Care has been taken to keep the physical prototype as simple as possible using low cost commercially available microcontrollers and locally available components to ensure adherence with the University budget. It eliminates unnecessary and redundant microcontrollers, reducing hardware costs without compromising on speed of sorting as it undercuts the commercial industrial alternatives.

9. Alignment with the Sustainable Development Goals

SDG 2 (Zero Hunger): Reduces food waste during post harvest operations by automatically detecting and isolating contaminated produce before post harvest distribution.

SDG 9 (Industry, Innovation, and Infrastructure): Implements affordable edge computing and sensor telemetry along agricultural value chains.

SDG 12 (Responsible Consumption and Production): Reduces waste in the production and supply chain and sets target quality criteria for trade in commercial agriculture.

10. Conclusion

The real-time computer vision and weight based produce sorting system meets all the structural, electrical and functional needs. The system features a dual-processor structure, multi-protocol communication, and separate user modularity, delivering a highly scalable and robust solution for agriculture sorting automation.